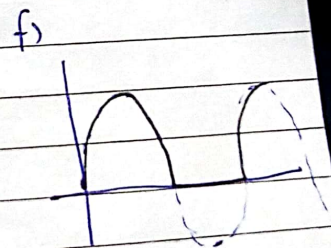
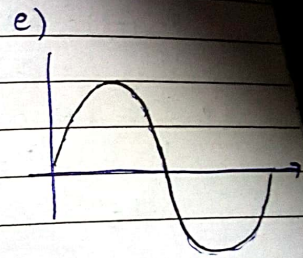
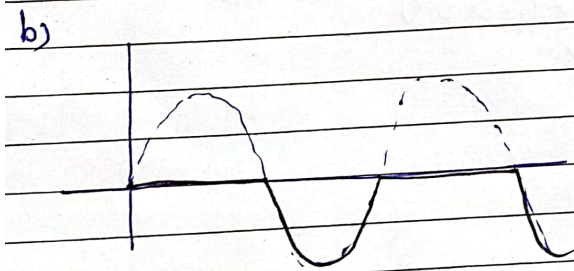
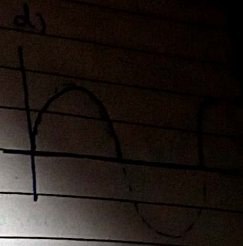
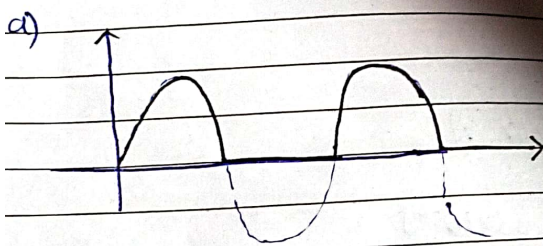
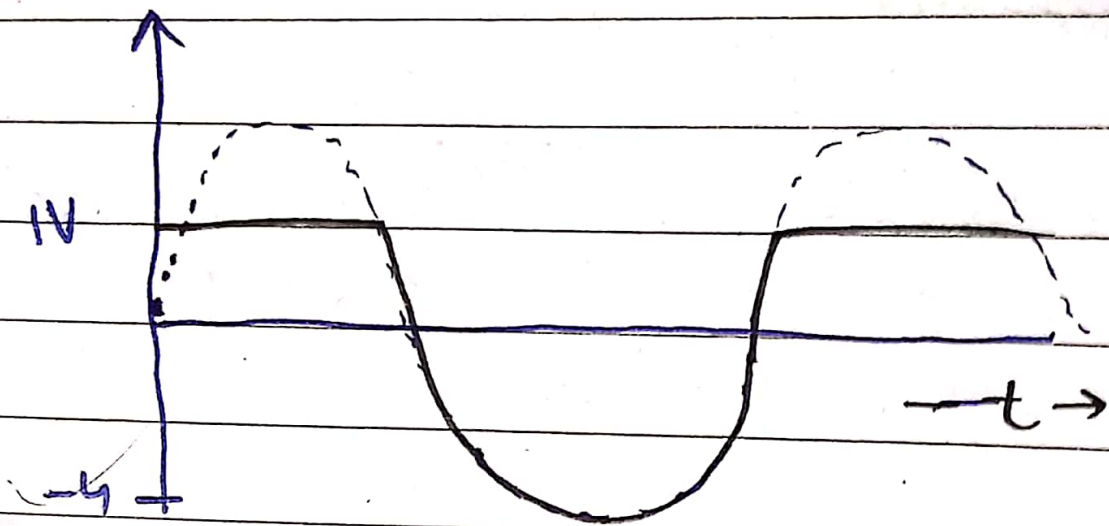


Name: Hureyah Hussain  
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Sect: BSEE-213-3A  
Assignment # 01  
(EdC Assignment)

Q#01-



Q#2)-



Q#2)-

$$\text{So, } I_s = 6.91 \times 10^{-8} \text{ A}$$

Now

$i_d$ , when

$$V_d = 0.71$$

$$i_d = 6.91 \times 10^{-16} e^{0.71 / 25 \times 10^{-3}} \\ = 1.49 \times 10^{-3} \text{ A}$$

$$V_d = 0.8 \text{ V}$$

$$i_d = 6.91 \times 10^{-16} e^{0.8 / 25 \times 10^{-3}} \\ = 0.0546 \text{ A}$$

$$V_d = 0.609 \text{ V}$$

$$i_d = 6.91 \times 10^{-16} e^{0.609 / 25 \times 10^{-3}} \\ = 6.7 \times 10^{-4} \text{ A}$$



Q#5p-

$$V_d = 0.7V$$

$$i_D = 1mA$$

$$\text{resistance of diode} = 0.7K\Omega$$

$$S_o = 0.7 \times 4$$

$$= 2.8K\Omega$$

$$I = \frac{3}{2.8 \times 10^3}$$

$$= 1.071 \times 10^{-3}A$$

$$R = \frac{16-3}{1.071 \times 10^{-3}} = 6.5 \times 10^3 \Omega = 6.5K\Omega$$

Q#6p-

$$I_{D1} = I_{S1} e^{\frac{V_{D1}}{V_T}}$$

$$I_{D2} = I_{S2} e^{\frac{V_{D2}}{V_T}}$$

$$I_{S1} = 10I_{S2}$$

$$I_{D1} = 10I_{S2}$$

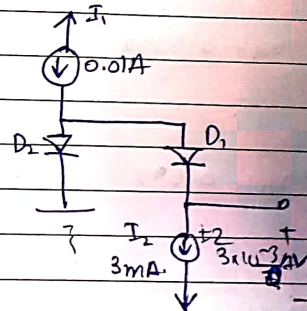
$$I_{D1} = 10I_{S2}$$

$$I_{D1} = 10I_{S2} e^{\frac{V_{D1}}{V_T}}$$

$$\frac{I_{D1}}{10I_{S2}} = e^{\frac{V_{D1}}{V_T}}$$

$$\frac{I_{D1}}{10I_{S2}} = e^{\frac{V_{D1}}{V_T}}$$

$$= 0.1 e^{\frac{V_{D2} - V_{D1}}{V_T}} \quad (1)$$



$$= 0.1 e^{\frac{V_{D2} - V_{D1}}{V_T}} \quad (1)$$

$$V_0 = V_{D2} - V_{D1}$$

$$= V_T \ln \left[ \frac{10 I_{D2}}{I_{D1}} \right] \quad (2)$$

$$I_1 = I_{D2} + I_1$$

$$I_{D2} = I_1 - I_{D1}$$

$$I_2 = 10^{-3}$$

$$= 7 \text{ mA}$$

$$V_0 = V_{D2} - V_{D1}$$

$$= 25 \times 10^{-3} \times \ln \left[ \frac{70}{3} \right]$$

$$V_0 = 0.074 \text{ V}$$

$$T_m = 1.0 \times 10^{-3} \text{ s}$$

3V

$$i_{D2} = \frac{V - V_D}{1000}$$

$$i_{D2} = i_{D1}$$

$$\frac{V - 0.7}{1000} = 1.01 \left( \frac{V - 0.6}{1000} \right)$$

$$V - 0.7 = 1.01 V - 0.606$$

$$-0.01 V = -0.004$$

$$V = -9.4 V$$

$$i_{D1} = \frac{V - V_D}{R}$$

$$= \frac{3 - 0.7}{1000}$$



Q#9)-

$$V_s = 0.7$$

$$V_s = \sqrt{2} V_{rms}$$

$$= \sqrt{2} \times 120$$

$$= 169.7 \text{ V}$$

$$S = V_s$$

Q#1030-

a)

$$V_{avg} = 10V$$

$$V_{in} = 169.7V_{sin}$$

$$V_2 = 0.7V$$

$$V_{avg} = 0.636V_s$$

$$V_s = \frac{10}{0.636}$$

$$= 15.7V$$

Transformer is centre tapped,

$$V_{ss} = 1.7 \times 3$$

$$= 31.4V$$

$$\frac{V_{ss}}{V_p} = \frac{N_2}{N_1}$$

$$\frac{31.4}{169.7} = 0.185$$

0.185 is the turn ratio

b)

$$V_{avg} = 100V$$

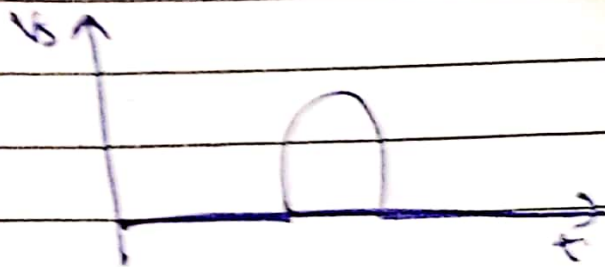
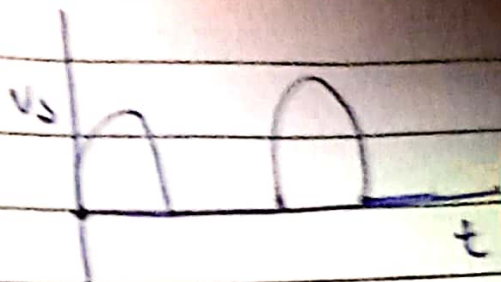
$$\frac{N_2}{N_1} =$$

turn ratio  $\times 10$

$$= 0.185 \times 10$$

$$= 1.85$$





$$0.636 V_s = V_{avg}$$

$$V_s = \frac{1s}{0.636}$$

$$V_s = 23.6V$$

$$V_p = 23.6 + 2$$

$$= 47.2V$$

$$P_i V = V_s - V_0$$

$$= 23 - 0.7$$

$$= 22.9V$$

$$V = 17.8$$

C) B -

